DEMO BRIEFING · CONFIDENTIAL

Proton

The patient app, the clinic console, and the one database both are actually writing to. Everything you need to explain it — including what it does not do yet.

Version 1.0 · pilot cut **Date** 28 Oct 2026 **Claims** docs/VALIDATION.md **Data** synthetic only

THE 30-SECOND PITCH

"Proton is the clinical record both sides can actually see. A patient on an Android phone books a visit, answers a check-in, asks for a refill, or uploads a document — offline if she has no signal. The moment the phone reaches the network, it writes to the same Postgres rows the doctor's console is reading. There is no integration, no nightly sync, no second copy. The clinician side is an AI copilot on top of that record: it drafts the note, checks the prescription, and never signs anything. The doctor signs."

If they remember one line: *"Two apps, one database."* Everything after that is detail.

01 The problem

	TODAY, WITHOUT PROTON
Patient	A portal she opens twice a year. Booking means a phone call in office hours. A question after a visit has nowhere to go. Records are PDFs in an email.
Clinic	Front desk retypes what the patient already typed. Messages land in a voicemail box detached from the chart. The doctor retypes the story into a note the patient will never see.

The record	Fragmented across the portal, the fax, the EHR and the patient's memory. Nothing reconciles.

Proton's bet: **the patient is a first-class writer to the clinical record**, and the clinician's job is to verify — not to transcribe.

02 The shape of the system

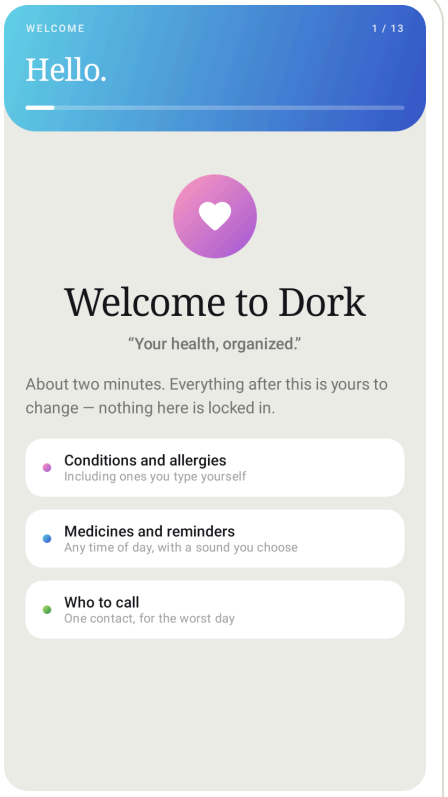
LAYER	WHAT IT IS
Patient app	Native Android · Kotlin + Jetpack Compose · offline-first outbox
Clinic console	React 19 + Vite + TypeScript + Tailwind · Care Loop polls every 3 s
API	Node 20+ / Express · deterministic safety rules · AI provider gateway
Operational store	PostgreSQL 16 · workflow state, outbox, audit chain, idempotency, tenancy
Clinical record	Medplum FHIR R4 · OAuth2 PKCE · standards vocabulary
AI	Gemini / Grok / DeepSeek / deterministic demo mode — server-side only

FHIR R4 is the vocabulary; Postgres is the operational truth. They meet in the API, not in the browser. And the AI never sits in the write path of a clinical decision: it proposes, a physician attests.

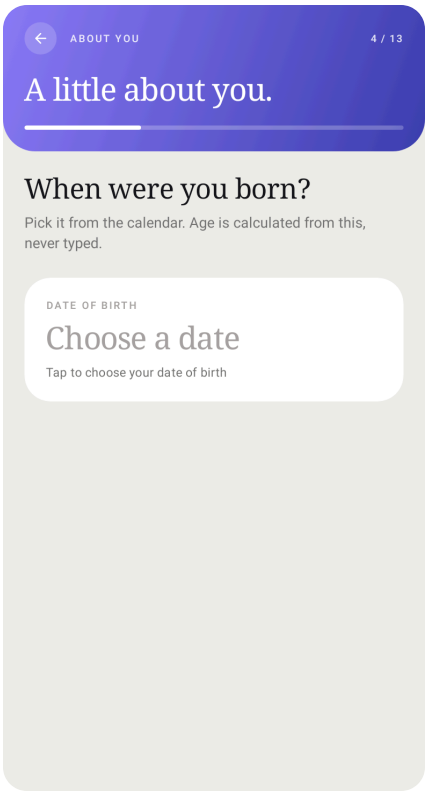
03 The mobile app

Not a web view. A first-party Compose app with its own design system, its own persistence, and its own sync engine.

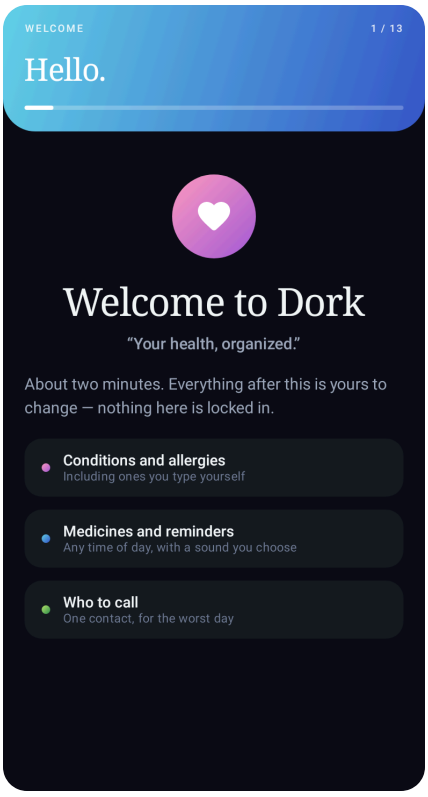
Name on device	**Dork** — working name, isolated to presentation strings; rebrand target is *Proton Patient*
Stack	Kotlin · Jetpack Compose (Material 3) · Gradle 9.1 · JDK 17
Package / version	`health.dork.app` · 1.0 (versionCode 1)
Android range	minSdk 26 (Android 8.0) · compile/target 36 · ~22 MB debug APK
Code	119 Kotlin files · ~21,800 lines · 28 test files / ~3,900 lines
Session	30-day patient session after a one-time sign-in code



Welcome · 1 of 13



Date of birth · 4 of 13



Real dark mode

The design language — first thing they will feel

- **Warm paper, not clinical white.** Canvas `#EDECEA`, borderless white cards, hairline borders. Colour is reserved for clinical signal — critical, warning, confirmed — never decoration.
- **Editorial type.** Serif display with tight tracking, sans body, serif-italic card labels. Numbers that matter are drawn with a dot-matrix component, not typeset.
- **Real dark mode.** Deep ink-blue `#0B0E14` with three visible elevations — not inverted paper, not pure black that eats a 2 px border on OLED.
- **One continuous surface.** A floating pill tab bar: five icons; the active one grows into a dark pill and reveals its label.
- **Onboarding gets colour on purpose.** A deterministic gradient band and progress line per step, because a medical intake that looks like a tax form is an intake people abandon — often in a waiting room, at night, worried.

The 13-step onboarding

Progressive, validated per step, resumable, and every answer editable later. Order: welcome → account → full name → **date of birth** (picked from a calendar; age is derived, never typed — leap-year and timezone safe) → sex → height → weight → blood group → conditions → allergies (free-text supported) → medications → emergency contact → complete.

The intake is not a formality: conditions and allergies entered here are exactly what the clinician's safety rules check prescriptions against later.

The five tabs

HOME	Not a feed. A context-aware card stack: next appointment and its status, next doses with *Mark taken* / *Skip*, a vitals snapshot, recent records, and one **next action** card that changes with state — complete intake, answer a check-in, request a refill, book a visit. Nothing is more than one tap from the answer.
CARE	Upcoming appointments (confirm / cancel / reschedule), doctor discovery by specialty and clinic, real availability, booking (slot → reason → confirm), pre-consultation intake (complaint, duration, severity 1–10, symptom checklist), post-visit check-in (better / same / worse, side effects, need to be seen), refill requests, and **From your clinic** — replies the doctor actually wrote.
RECORDS	A chronological timeline uniting consultations, labs, prescriptions and imaging; record detail with its source document; prescription detail with the signed order; document upload from camera or file that is **sent to the clinic**, not just stored on the phone.

DORK AI	“Your health information, explained.” Answers cite the patient’s own records. Urgent symptoms are escalated immediately instead of answered — it is explicitly not a diagnosis channel, and the copy says so.
PROFILE	Health profile (editable), **Medical ID** for first responders (blood group, severe allergies, active medications, one-tap emergency dial), preferences, clinic connection, sync controls, sign out, and **Erase all data** — a device wipe, not a logout.

Plus 17 sub-screens reached from the tabs: doctor detail · book · reschedule · intake · follow-up · refill · record detail · prescription detail · document upload · doctor chat · clinic inbox · health profile · medical ID · medications · clinic · sync · care packet.

Offline — the part worth demoing

The phone is not “cached”. It is **durable**:

1. The action is written to a local **Room-backed outbox** with a stable **idempotency key** for that row.
2. `WorkManager` (plus a foreground drain while the app is open) retries with backoff. Airplane mode is just “retry later”.
3. On success the server returns the canonical row; local state reconciles. Replaying the same key returns the original — no duplicates.
4. Pull is caller-scoped: the patient sees their own rows, derived from real columns (`confirmed_at` → `CONFIRMED`), not a mirrored copy.

Consequence on stage: **booking in airplane mode is not a trick.** The row simply has not left the device yet. When signal returns it lands exactly once.

Identity, security, privacy

- **Clinic link, code + one-time code.** Clinic code → email → 6-digit code → 30-day session. Identity always comes from the issued session, never the request body.
- **Tenant isolation.** A second clinic sees an empty schedule and cannot act on the first clinic’s rows.
- **PHI never in logs.** Masking helpers; emails render as `c***@example.com`.
- `FLAG_SECURE` on high-sensitivity views — screenshots and screen recording blocked where it matters.
- **Erase all data** wipes local state, session and outbox.

04 The clinic side — the laptop half

Care Loop is the live console, and the only clinic view reading the same rows the phone writes: today's schedule with confirmation state (3 s poll), a **Needs attention** queue sorted by clinical urgency rather than arrival, reply threads, patient documents and signed prescriptions in one sheet. Front desk can confirm; only a clinician can author a clinical reply.

Around it: the consultation cockpit (ambient capture → SOAP draft with provenance badges → deterministic prescription checks → PDF), document review with fact-level approve/reject and citations, the patient profile and longitudinal timeline, and self-serve clinic onboarding.

> **DO NOT IMPLY**
>
> The older **Schedule** and **Patients** pages read a seeded workspace. Only Care Loop is the live loop. Never show the two side by side as if they were the same data.

05 Numbers that are true

119	51	66	27
Kotlin files · ~21.8k lines	API suites · 8 need Postgres	workflow + live-loop checks	tables · 9 migrations

Also true: 5 tabs · 17 sub-screens · 13 onboarding steps · 28 Android test files. Do **not** quote "212 tests", "100/100", "35/35 dimensions" or "clinically validated" — those live in historical milestone reports and are explicitly retracted in `docs/README.md`.

06 Real vs. not real — say it before they ask

proven

- Airplane-mode booking lands in Postgres once and appears on the console within ~3 s of reconnection.
- Clinic confirmation reaches the phone on its next pull, from the same column.
- A check-in becomes a severity-sorted attention item (critical above routine).

not yet

- No public deployment. Today's demo is **localhost on the laptop, phone over USB** (`adb reverse`). An older deployed API host 404s every product route — never show or reuse it.
- No push notifications. FCM is next; the phone polls every 15 s while open, and reminders are local exact alarms. A closed app does not light up.

- A clinician reply reaches the patient inbox; the patient's own message is never echoed back as clinic mail.
- Tenant isolation, role gating, idempotent replay, audit-chain tamper detection, state and outbox survival across an API restart, dump → restore.
- Production image builds, boots in production posture, and fails closed on the real routes.
- No live-provider AI evaluation, no clinician-observed study, no load / pen / accessibility testing, no real Medplum round-trip in CI.
- Synthetic patients only. No BAA, no real PHI.
- The clinical agent stack is not yet bound to physician attestation — internal, not deployable.

07 The 3-minute demo

Setup: `npm run demo:seed` (prints the clinic code and logins) → API on `:3002` → web on `:3001` → Care Loop → phone via `adb reverse` → sign in once. Full pre-flight and failure recovery: `docs/DEMO.md`.

40 s

1 · She books with no signal

Phone in airplane mode → **Care** → **Book appointment** → any slot → "Sore throat" → Confirm. Hold the phone up so the airplane icon is visible.

"There is no network on this phone. The booking is on the device — watch the laptop: it genuinely doesn't have this visit."

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15 s

2 · The clinic sees nothing yet

Laptop: Care Loop has the seeded visits and the refill request — not the new booking.

"Same console a real front desk uses. It has no idea that booking exists, because it doesn't. That is the honest state."

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20 s

3 · Signal comes back

Airplane mode off, app in foreground. Within 3 s the visit appears as "Awaiting confirmation".

"One request went out. The clinic's poll picked it up. Nothing was re-keyed."

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20 s

4 · Reception confirms

Click **Confirm**; the badge flips. The phone's next pull shows Confirmed.

"That wrote one column — confirmed_at — on one row. There is no second copy to drift out of sync."

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30 s

5 · She says it got worse

Phone: **After your last visits** → **Answer check-in** → Worse + side effects + needs to be seen. A critical card appears above the routine refill.

"The queue is sorted by clinical urgency, not by arrival. A patient asking to be seen never sits below a refill request."

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45 s

6 · The doctor answers

Laptop: **Open & reply** → type the reply → **Send reply**. Within 15 s the phone shows "From your clinic — 1 new message".

"That message is the row the doctor just wrote. The phone didn't get a push from a demo server — it pulled the clinic's own record."

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15 s

Close

"Two clients, one Postgres, no glue in between. A patient app that works offline, and a clinic console that sees the truth the moment the phone can reach it."

Reset between rehearsals: `npm run demo:reset` — deletes only `pat-demo-*` rows and issues a new clinic code; the phone keeps its session. Memorise three recoveries: API without `DATABASE_URL` (restart, no reseed) · cable pulled (replug + `adb reverse` + **Sync now**) · stale tab (click **Refresh**).

08 Questions they will ask

"Is this HIPAA compliant?"

No — and say so plainly. This is a synthetic-data build with no BAA. HIPAA readiness is a deployment posture: BAAs, tenant boundaries, KMS encryption, audit retention. `docs/trust/` describes exactly what would need to be true.

"Is the AI making clinical decisions?"

No. It drafts and it checks. Deterministic rules refuse an unsafe prescription — missing dose, frequency or duration; duplicate orders; cross-reactive allergies. Nothing becomes a signed note or order without physician attestation.

"What happens if the network dies mid-send?"

The row stays in the outbox with its idempotency key and retries. A replay returns the original appointment instead of creating a second one. Tested, including process death and an API restart.

“How is this different from a patient portal?”

A portal reads the record. This writes to it — in the same schema the clinic reads — and reconciles deterministically. There is no second copy to drift.

“Can I see it on my phone?”

Today: install the APK and point it at the laptop over USB. A hosted URL is the pilot cut’s Days 1–2 work — do not promise it on the spot.

“What’s missing before a real clinic?”

Deployment, FCM push, a restore drill on real hardware, a named clinician owner, and a signed BAA — in that order.

“Why does the icon say Dork?”

Working name, isolated to presentation strings. Domain models, repositories and FHIR resources are already neutral, so the rebrand is a rename, not a rewrite.

09 Words to use and avoid

SAY	“Durable offline outbox.” “One database, both halves.” “It proposes; the physician attests.” “Synthetic cohort.” “Hermetic regression suites.”
DON’T SAY	“HIPAA compliant.” “Clinically validated.” “AI diagnosis.” “Real-time push.” “Deployed and live.” “Production-ready.”
IF UNSURE	“I can show you exactly what the tests cover — it’s docs/VALIDATION.md.” That sentence wins more trust than any number.

Companion documents · DEMO.md (stage script and recovery) · VALIDATION.md (claim boundaries) · PILOT_CUT.md (what’s left before a pilot) · Android architecture.

Proton · internal demo briefing · synthetic data only.